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جامعة الملك عبدالعزيز

ACADIFY
Academic Simplify

Acadify: Academic Simplify

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CPIS 498 Senior Project

Introduction

The Acadify system is developed using the ASP.NET MVC architecture. This architecture divides the system into three main parts: Model, View, and Controller. The Model handles the data and business logic. The View shows the user interface. The Controller connects the Model and the View. This structure makes the system easier to manage, update, and maintain.

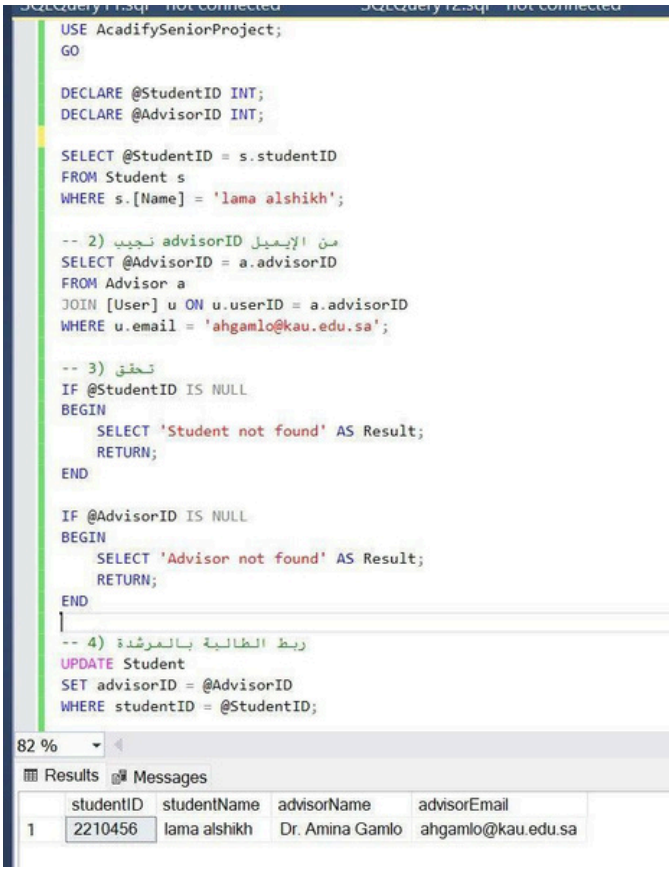
The system is developed using Microsoft Visual Studio. It is used to build and manage the ASP.NET MVC application. For storing and managing data, Microsoft SQL Server is used as the database management system. SQL Server Management Studio (SSMS) is used to create the database, design tables, create views, and run queries. The database stores all academic data, such as student records, advisor information, transcripts, graduation status, and matching status.

Acadify also includes an intelligent rule-based agent inside the system environment. This agent analyzes academic data, such as completed courses, remaining requirements, credit hours, and graduation status. Based on predefined academic rules, the agent helps in course recommendations, student status evaluation, and automatic advising form processing. The agent does not use machine learning models. Instead, it follows fixed academic rules to provide accurate and consistent decision support.

This system environment provides a stable, secure, and scalable platform. It supports academic advising processes and ensures reliable system performance.

1-SQL Environment Setup

Microsoft SQL Server was installed and configured as the main database system.SQL Server Management Studio (SSMS) was used to create the database, tables, and relationships.The Acadify database was set up to store student, advisor, transcript, graduation, and matching status data.



```
USE AcadifySeniorProject;
GO

DECLARE @StudentID INT;
DECLARE @AdvisorID INT;

SELECT @StudentID = s.studentID
FROM Student s
WHERE s.[Name] = 'lama alshikh';

-- 2) نجيب advisorID من الإيميل
SELECT @AdvisorID = a.advisorID
FROM Advisor a
JOIN [User] u ON u.userID = a.advisorID
WHERE u.email = 'ahgamlo@kau.edu.sa';

-- 3) تحقق
IF @StudentID IS NULL
BEGIN
    SELECT 'Student not found' AS Result;
    RETURN;
END

IF @AdvisorID IS NULL
BEGIN
    SELECT 'Advisor not found' AS Result;
    RETURN;
END

-- 4) ربط الطالبة بالمرشدة
UPDATE Student
SET advisorID = @AdvisorID
WHERE studentID = @StudentID;
```

82 %

Results | Messages

	studentID	studentName	advisorName	advisorEmail
1	2210456	lama alshikh	Dr. Amina Gamlo	ahgamlo@kau.edu.sa

Figure 1: Advisor Assignment

This figure shows assigning a student to an academic advisor using the advisor’s email.

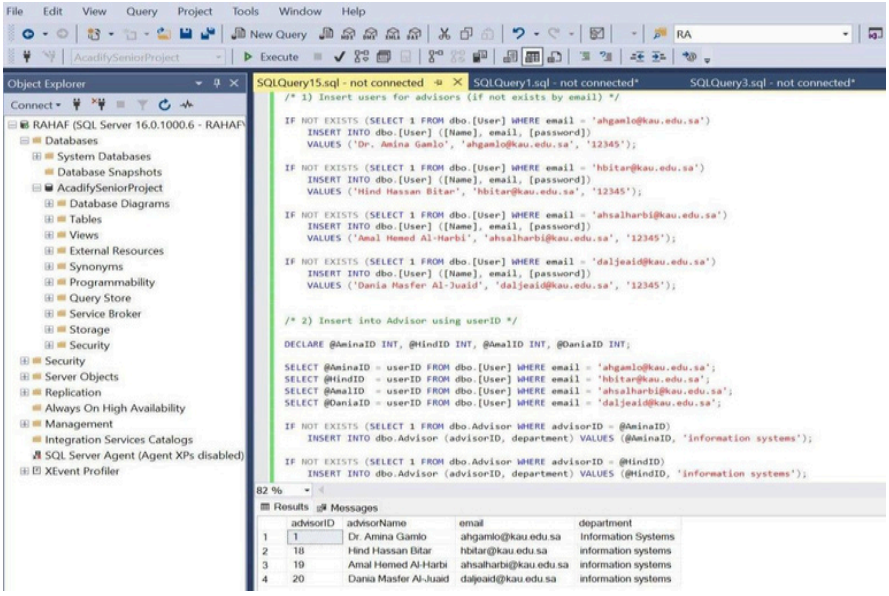


Figure 2: Advisor Accounts Creation

This figure shows inserting advisor user accounts into the system database and linking them with the Information Systems department.

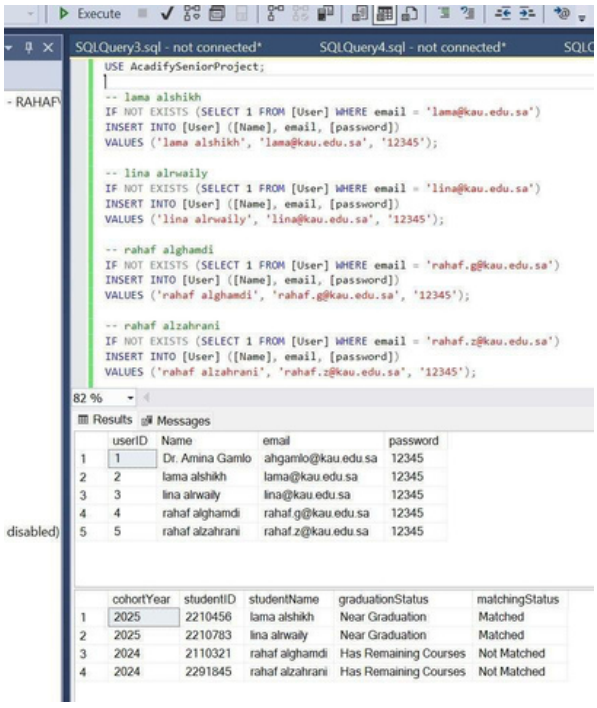


Figure 3: Student User Accounts

Thisfigureshowscreating student user accounts with basic login information.

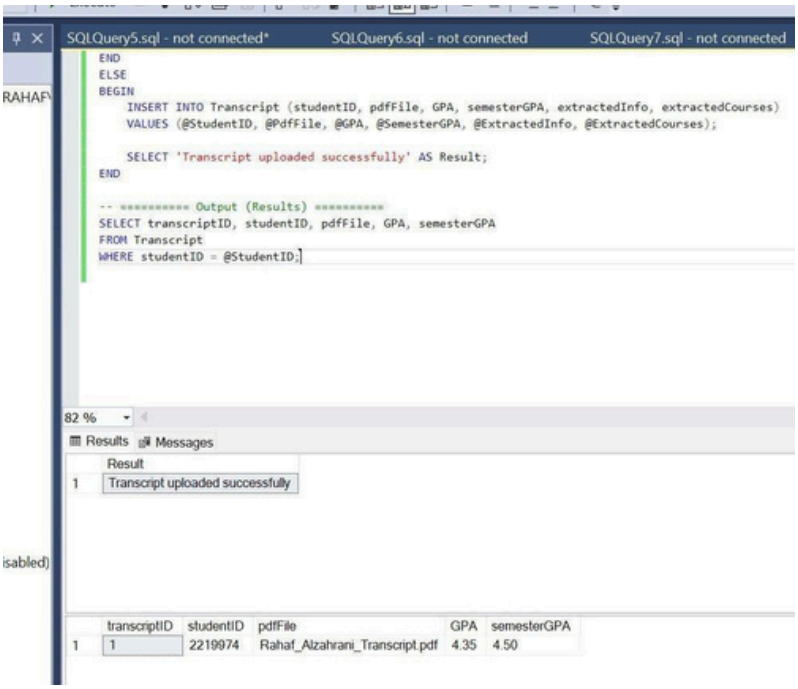


Figure 4: Transcript Data Storage

Thisfigureconfirmsthat the transcript data is stored successfully in the database.

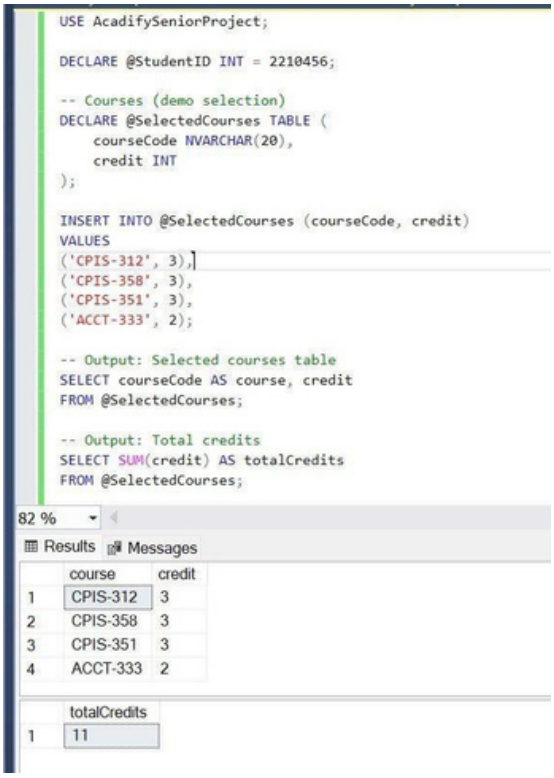


Figure 5: Selected Courses and Credit Calculation

This figure demonstrates selected courses and calculates the total registered credit hours.

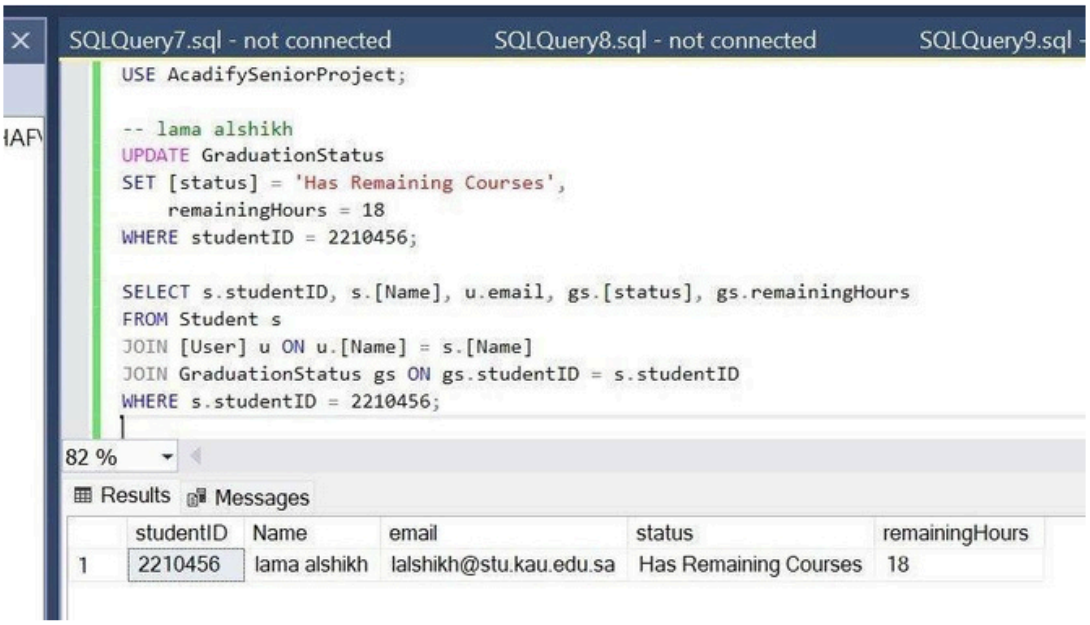
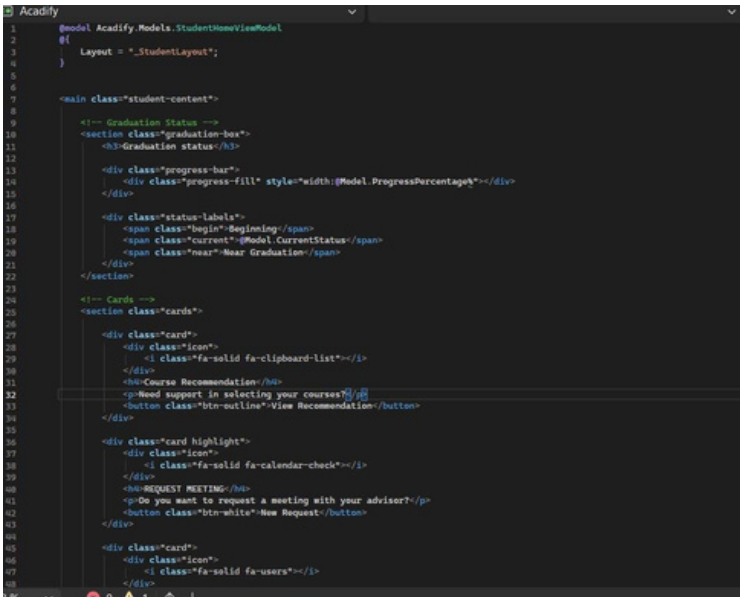


Figure 6: GraduationStatus Management

This figure shows updating the student graduation status and remaining credit hours.

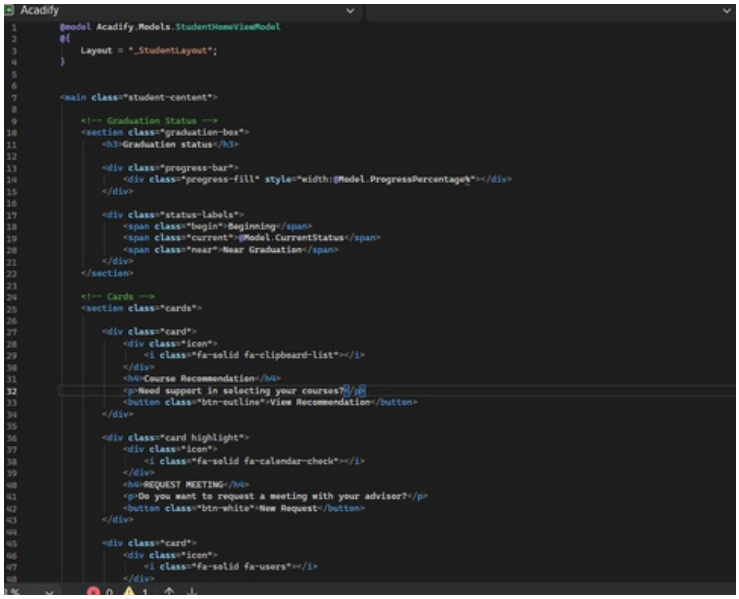
2-Visual studio setup

Microsoft Visual Studio is the main development environment used to build the Acadify system. It is used to create and manage the ASP.NET MVC project, including the Models, Views, and Controllers. The system uses C# for the main logic, HTML and CSS for the user interface design, JavaScript for simple interactive features, and SQL for database management in Microsoft SQL Server. Visual Studio helps in writing code, organizing files, debugging errors, and connecting the system to the database. Overall, it provides a clear and organized setup that supports system development and testing.



StudentHome View :Figure 7

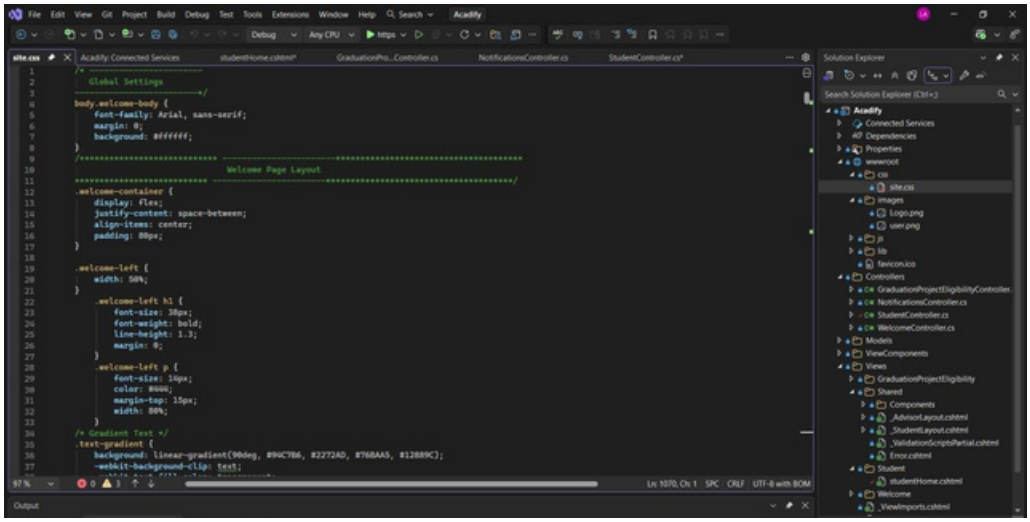
this code creates a simple dashboard where the student can see her academic progress and access important services quickly.



```
Acadify
@model Acadify.Models.StudentHomeViewModel
@{
    Layout = "_StudentLayout";
}


<main class="student-content">

    <!-- Graduation Status -->
    <section class="graduation-box">
        <h3>Graduation status</h3>

        <div class="progress-bar">
            <div class="progress-fill" style="width:@Model.ProgressPercentage%"></div>
        </div>

        <div class="status-labels">
            <span class="begin">Beginning</span>
            <span class="current">@Model.CurrentStatus</span>
            <span class="near">Near Graduation</span>
        </div>
    </section>

    <!-- Cards -->
    <section class="cards">

        <div class="card">
            <div class="icon">
                <i class="fa-solid fa-clipboard-list"></i>
            </div>
            <h4>Course Recommendation</h4>
            <p>Need suppsert in selecting your courses?</p>
            <button class="btn-outline">View Recommendation</button>
        </div>

        <div class="card highlight">
            <div class="icon">
                <i class="fa-solid fa-calendar-check"></i>
            </div>
            <h4>REQUEST MEETING</h4>
            <p>Do you want to request a meeting with your advisor?</p>
            <button class="btn-white">New Request</button>
        </div>

        <div class="card">
            <div class="icon">
                <i class="fa-solid fa-users"></i>
            </div>
```

StudentController :Figure 8

this controller prepares student data and decides the graduation status based on the progress



```
/*
   Global Settings
-----------------------*/
body.welcome-body {
    font-family: Arial, sans-serif;
    margin: 0;
    background: #ffffff;
}
/************************* ----------------------------------------------------
                          Welcome Page Layout
************************* ----------------------------------------------------*/
.welcome-container {
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 80px;
}

.welcome-left {
    width: 50%;
}
    .welcome-left h1 {
        font-size: 38px;
        font-weight: bold;
        line-height: 1.3;
        margin: 0;
    }
    .welcome-left p {
        font-size: 16px;
        color: #666;
        margin-top: 15px;
        width: 80%;
    }
/* Gradient Text */
.text-gradient {
    background: linear-gradient(90deg, #9AC7B6, #2272AD, #768AA5, #12889C);
    -webkit-background-clip: text;
```

Figure 9: site.css

this CSS file controls the visual design of the Welcome page and general system appearance.

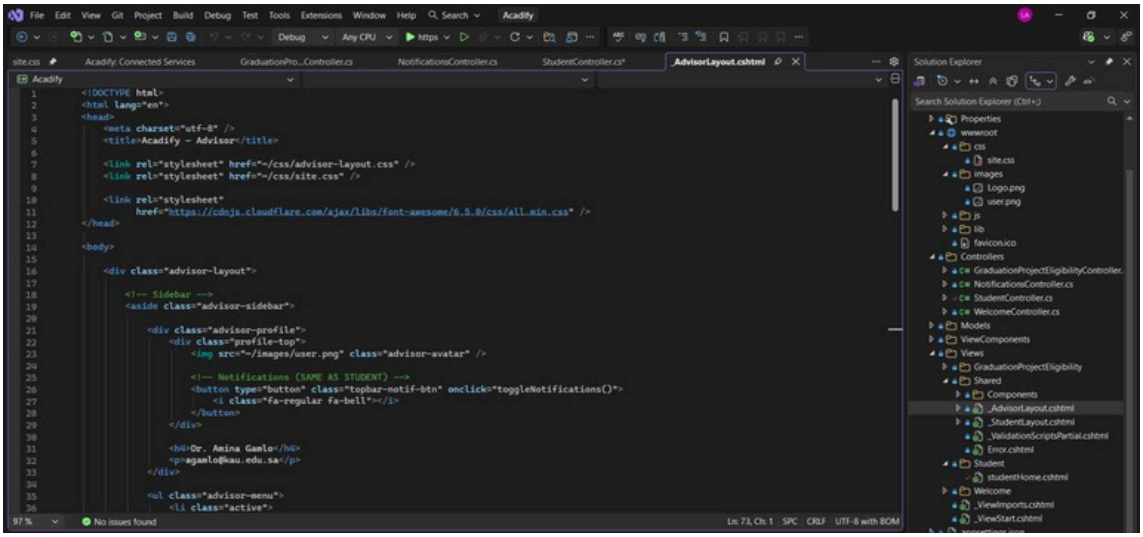
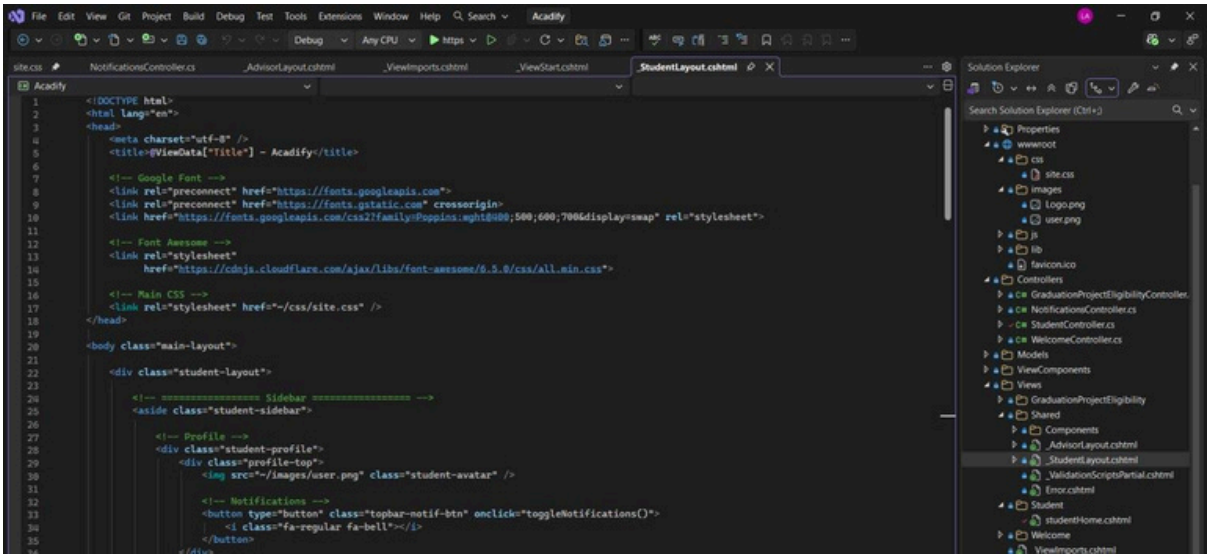


Figure 10: StudentLayout

profile ,It includes the sidebar .this layout buildsthe structure for all student pages and main content area ,notifications ,area



AdvisorLayout :Figure 11

this layout builds the advisor dashboard structure. It includes the sidebar, profile info, notifications, and navigation menu.

3- interfaces

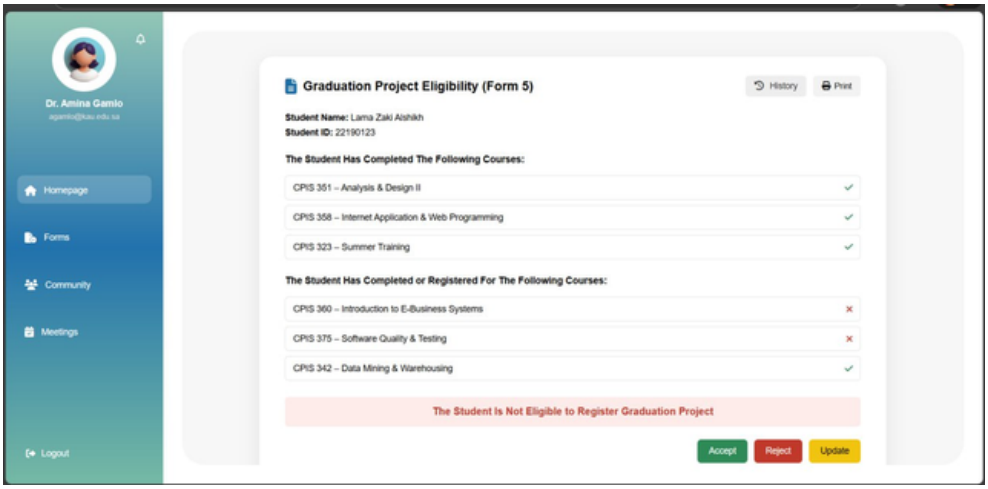


Figure12: form 5 interface

this page verifies graduation project requirements and shows whether the student is eligible or not.

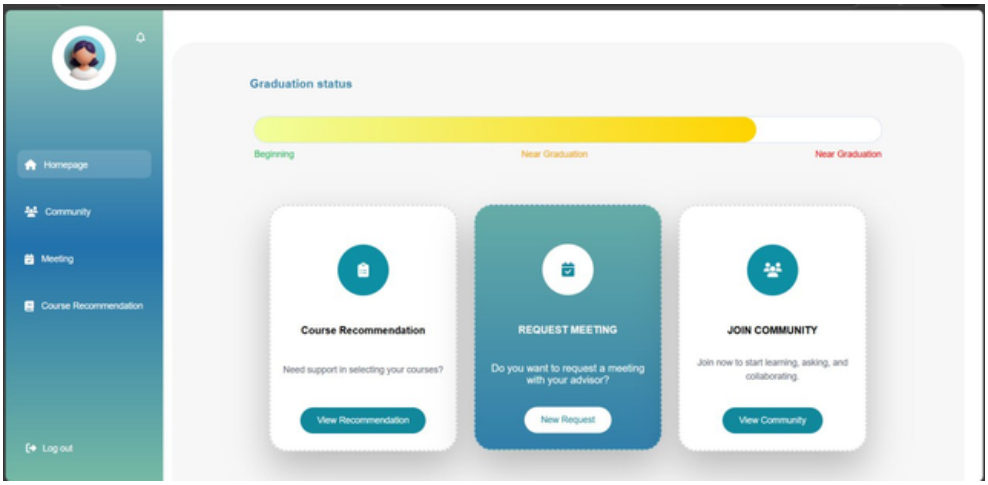


Figure13: student homepage

this page gives the student a clear view of academic progress and quick access to important services. "student homepage"

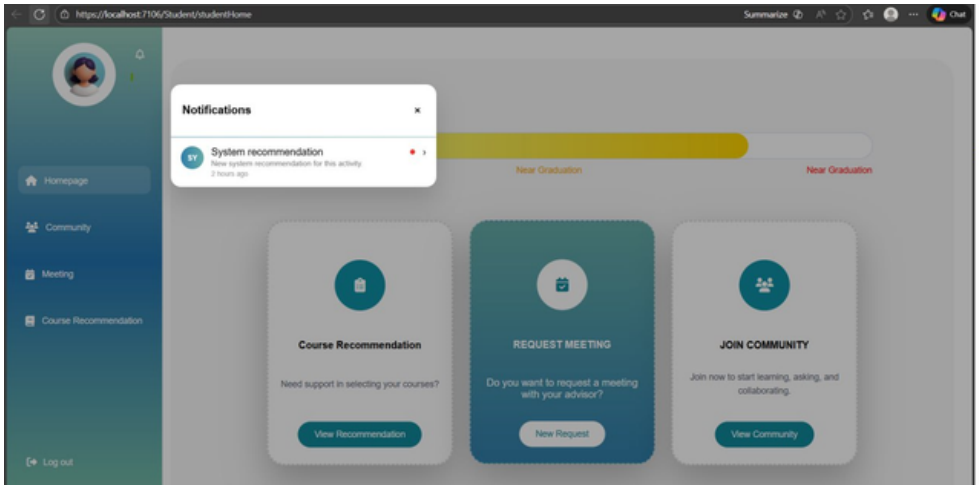


Figure14 : notification

This page allows the student to quickly view new updates and notifications. "Notifications Panel"